

THIS IS A COMPLETELY NEW PROJECT PROPOSAL

Revised Project Proposal

Name: Kyu Lee

UNI: kbl2140

For my final project, I would like to build:

A Chrome extension that will provide real-time public opinion estimates using LLM-based synthetic surveys based on selected social media content (tweets, reddit posts, YouTube comments, etc.). The tool is called EchoBreaker.

The Problem:

Studies have shown those who post on social media are not representative of the general population in regards to demographic composition, viewpoints, and emotional intensity. However, due to the large volume of social media consumption, users often have a distorted view of what most other people think. This contributes to polarization, and can influence their own feelings and decisions. EchoBreaker is a real-time calibration tool that helps people understand how much of what they see actually reflects public opinion. The goal is to foster a more accurate understanding across differences of opinions.

This project/paper will make three contributions:

1. EchoBreaker as a research artifact. An example of a high level user-flow is this: user browses Twitter, becomes interested in a post, clicks on it to trigger EchoBreaker extension, EchoBreaker runs synthetic survey on post (users can continue browsing), EchoBreaker shares results of its analysis.
2. Findings from a think-aloud study with 5 real twitter users using EchoBreaker. Each participant will browse real Twitter threads (will be pre-generated) on 3 topics. For each topic, they will first articulate what they think public opinion is based on the thread. Then I will ask them to use EchoBreaker to see its output. Then I will observe and record their reaction (are they surprised? Skeptical? etc.). Afterwards, I'd conduct a semi-structured interview covering trust, usefulness, product feedback, and whether EchoBreaker changed their minds on what they just read. The main question I'm trying to answer with this study is: "how do users interpret EchoBreaker's output given that it disagrees with social media posts? Would this reduce their own perception gaps?"
3. Validation study where we take topics where ground truth exists (like polling) and compare EchoBreaker's synthetic survey estimate against that truth. I would pick 3 topics where high-quality polling data exists with demographic breakdowns. Then I would compare it with results from EchoBreaker's analysis.

Note: My thought is that for a publishable paper, my think-aloud study + validation study is not “good enough” but is within reasonable scope for this class. Probably what is more interesting to publish is a quantitative user survey where I measure perception gaps, confidence, and attitude shifts with a large number of participants (~100 or more). I do think I will get interesting learnings from this think-aloud study that can help me with the longer survey design which I can pursue even after 6156 ends.

Disclaimer: I have already asked for advice from Dr. Brian Smith and also might talk to other professors about this project. However, I am not working on this project as part of any other class. I am also not working on it with any other students.